Towards Lifelong Learning of Object-Oriented Preferences for Personalized Robotic Assistance

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Abstract: For safe and fluent human-robot interaction (HRI), robotic assistants must not only complete tasks but do so while respecting their user's preferences. In lifelong deployment scenarios, these preferences are often not all readily apparent and may surface in different situations. We study how a robot can continually learn these preferences while minimizing the total effort spent by the user in communicating them. First, we formalize the problem of *lifelong learning of object-oriented preferences* (LOOP), where a robot assists the user over a sequence of tasks and learns how objects should be handled from corrections. Next, we present CO-OPT, a continual learning approach that estimates user preferences about objects from language corrections to generate user-aligned plans, and apply it to household cleanup tasks with a mobile manipulator. In simulation experiments, CO-OPT reduces the corrections needed to align robot behavior. These results indicate that transferring learned preferences across tasks and objects lowers user burden, a prerequisite for robots to remain useful in long-term HRI.

Keywords: Lifelong Learning, Preference Learning, Learning from Corrections

1 Introduction

Across many domains, robots are increasingly envisioned as general-purpose assistants for human users [1]. In homes, for instance, open-vocabulary mobile manipulation (OVMM) lets robots interact with a wide range of everyday objects in previously unseen environments [2]. Capability alone, however, does not make a robot useful to a particular person. Most assistance tasks can be completed in many different ways, but the "correct" way depends on the specific user. A robot misaligned with these preferences forces the user to supervise and correct it, the very burden an assistant should remove. Thus, the robot must be able to align its behavior to its user during deployment.

More challenging is that preferences often differ across users and surface only when a situation demands them. In this work, we focus on *object-oriented* preferences. A robot interacts with many different objects over its lifetime, and a user may have preferences about how each of them is handled [3, 4]. In the cleanup scenario of Fig. 1, a user who wants the robot to avoid valuable electronics tells it not to clean up the cell phone; the same user may also care where certain objects belong and in what order they are put away. Ideally, a robot that learns from a correction should *generalize* it to similar objects, such as a camera in our scenario.

Since the user's preferences are latent, the robot must learn them from the user's inputs. These inputs can take many forms: demonstrations, comparisons, physical corrections, and natural language utterances [5, 6, 7]. We focus on corrections given in natural language [8, 9, 10, 11], which arise naturally when the robot violates a preference during a task. Irrespective of their form, user inputs should be kept to a minimum. Enumerating all of one's preferences up front is costly [12], and cognitive science suggests that people are often unaware of a preference until a situation demands it [13]. Thus, new preferences continue to surface as corrections throughout robot deployment that impose a cognitive and physical burden on the user, which a useful assistant should aim to minimize.

Workshop on Learning from Corrections and Interventions at CoRL 2026, Austin TX, USA.

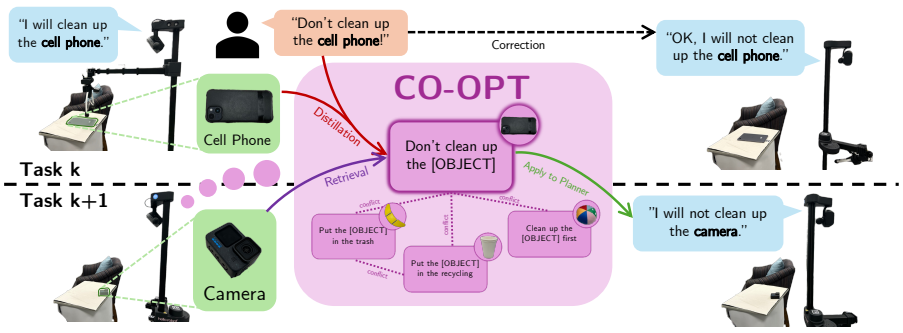


Figure 1: CO-OPT (**C**ontinual **O**bject-**O**riented **P**reference Learning in Situated **T**asks) is a framework for lifelong preference learning in long-term robot deployment. When the user provides corrections, CO-OPT generalizes them and stores them in its memory. To reduce the user's feedback effort for a new object or task, CO-OPT estimates their preferences by retrieving relevant stored corrections.

Reducing correction occurrences over time requires transferring what a correction teaches to situations not yet encountered. The lifelong learning framework [14] describes this ability for agents learning a sequence of tasks. In robotics, it has been explored extensively for learning how to *accomplish* tasks [15, 16, 17, 18, 19, 20, 21]. However, learning how to align a robot's task execution according to user preferences has received less attention in the lifelong setting. Preference learning in robotics mostly operates on a single task [22, 23, 24, 25]. Related are methods for long-term personalization in HRI. While these methods adapt user models over repeated sessions, they do not learn transferable preferences from corrections [26, 27, 28]. The regime in which preferences must transfer across an open-ended sequence of tasks and objects remains underexplored. We call this problem *lifelong learning of object-oriented preferences* (LOOP). To tackle this problem, this paper makes three main contributions:

- We formalize LOOP (Sec. 2) in which given the corrections received on earlier tasks, the robot must complete each new task with as few corrections as possible.
- As a solution to LOOP, we present CO-OPT (Sec. 3), a continual learning approach that estimates user preferences about objects from natural language corrections and uses them to generate user-aligned plans.
- We evaluate CO-OPT over long-term robot deployment in a simulated room cleanup scenario and show that it reduces the corrections users must provide.

Together, these results indicate that transferring preferences across *tasks* and *objects* lowers the user's burden, a prerequisite for robots to remain useful in long-term HRI.

2 Problem Formulation

We consider HRI settings in which a robot assists a single user over an open-ended sequence of tasks. The robot begins with incomplete knowledge of the user's preferences, so its initial behavior is misaligned with how the user wants tasks to be done. In response, the user may correct the robot and teach their preferences. To realize fluent HRI, however, the robot should require progressively fewer corrections as it gains experience. We call this problem *lifelong learning of object-oriented preferences* (LOOP) and formalize it next.

2.1 HRI Setting

Task Sequence and Environment. Following the lifelong learning setting [14, 15], the robot encounters a sequence of K tasks $\{T_1, \ldots, T_K\}$ specified by the same user U. We model each task T_i as a Markov Decision Process (MDP), with shared state space $\mathcal{S}$, action space $\mathcal{A}$, and transition function $\mathcal{P}$, and task-specific initial state distribution $\mu_{0,i}$ and task reward R_i. Every state $s \in \mathcal{S}$ includes a set of objects $O(s)$, and different tasks generally involve different objects. We additionally consider

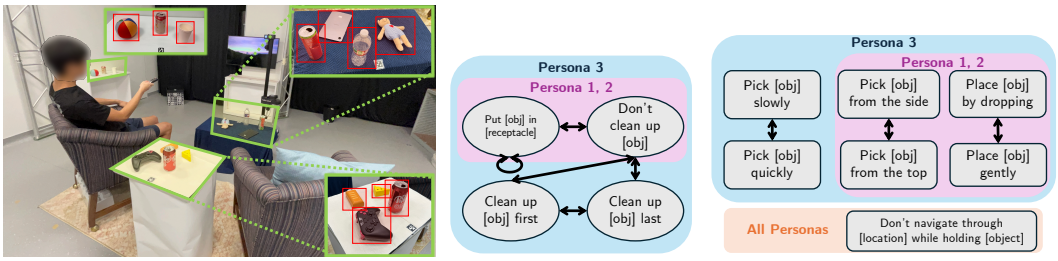


Figure 2: (left) A real-world instance of the household cleanup task, in which the highlighted objects must be tidied according to the user's preferences. (right) Object-oriented preferences in the household cleanup domain. High-level preferences are shown in ovals and low-level preferences in boxes. Edges connect preferences that conflict when applied to the same object, and all receptacle preferences conflict with one another. The labeled shaded areas denote the ground-truth preferences p_U enforced by the simulated personas (Sec. 4).

the sparse-reward setting: for a task execution trajectory $\tau \doteq (s_0, a_0, \ldots, s_H)$, the task reward reduces to the goal predicate $g_i(\tau) = \mathbb{I}[s_H \in \mathcal{G}_i]$, where $\mathcal{G}_i$ is the set of goal states for T_i.

User Preferences. Following preference learning methods [22, 23, 24], we model how well a trajectory τ aligns with the user's preferences through a *user preference reward* $\tilde{R}(\tau; T_i, p_U)$, where p_U denotes the underlying preferences for U. Both p_U and $\tilde{R}$ are unknown to the robot. LOOP focuses on *object-oriented* preferences involving how the robot should handle objects it encounters. To incorporate human preferences into planning, we follow computational frameworks in HRI [5] and decompose the robot's reasoning into a planner π and a *user model* ψ. The planner $\pi(\tau; T_i, p)$ produces trajectories for task T_i conditioned on a *preference estimate* p produced by the user model, described next.

User Model. The user model ψ outputs $p \doteq \psi(T_i)$, the robot's estimate of the user's true preferences p_U for a given task T_i. This preference estimate steers the planner toward some task solutions and away from others. The form of p depends on the planner: a reward term for RL [29], symbolic or motion constraints for TAMP [30, 31], or in-context examples for LLMs [3, 32]. For task T_i, we define p to be *feasible* if the planner can still complete the task under it; and *aligned* if it is both feasible and maximizes the expected user reward $\tilde{R}$. Solving LOOP amounts to estimating an aligned p on each new task with as little user feedback as possible.

User Feedback. We follow prior work on learning from language feedback [9, 10, 11] and let the user issue *natural language corrections* while watching the robot act. On task T_i the user provides a feedback set $F_i = \{\zeta_{i,1}, \zeta_{i,2}, \ldots\}$ of corrections, each triggered by a behavior that violates their preferences. We assume the user only issues a correction whenever the robot's behavior violates p_U. Thus, the number of corrections $C_U(\tau; T_i) = |F_i|$ the user gives while observing τ is an observable proxy for preference alignment; minimizing C_U is equivalent to maximizing $\tilde{R}$.

2.2 Problem Statement

In LOOP, after interacting with the user on tasks $T_1, \ldots, T_{k-1}$, the robot should complete the next task T_k while requiring as few corrections as possible. Since the planner π is given, the robot's behavior on T_k is determined by the user model ψ learned from the feedback sets $F_1, \ldots, F_{k-1}$. The objective is therefore to learn ψ that minimizes the expected feedback effort on the next task,

$$J_k(\psi) = \mathbb{E}_{\tau \sim \pi(\cdot; T_k, \psi(T_k))} \left[C_U(\tau; T_k) \right], \tag{1}$$

subject to the task being completed for every $k \geq 1$. By minimizing J_k for every k, we aim to minimize the total number of corrections $\sum_k J_k$ over the robot's lifetime.

2.3 Household Cleanup Domain

We illustrate LOOP in household cleanup (Fig. 2 left), the domain used in our experiments. Household assistance is a widely studied setting for robots [3, 2, 4], and it exercises every element of LOOP. The objects in a home vary widely and change from task to task, and users can hold many preferences about how their home is kept. The preferences we consider for this domain (Fig. 2 right), involve both high-level plan preferences and low-level task execution preferences.

3 Continual Object-Oriented Preference Learning in Situated Tasks

To optimize Eq. (1) from natural language corrections, we introduce the CO-OPT algorithm. CO-OPT parameterizes the user model ψ with a database V of feedback entries and learns through a distill-and-retrieve architecture. While this architecture is standard in lifelong learning with explicit knowledge [19, 21, 20], CO-OPT differs in what it stores and retrieves: object-agnostic templates anchored to concept-augmented representations of the corrected objects.

3.1 Feedback Distillation (Alg. 1)

Given a correction ζ on task T_i, CO-OPT *distills* it into a generalizable form and stores it in V.

Processing Feedback (Lines 1, 4–8). A correction is phrased in the user's own words about the object at hand. CO-OPT assumes that the preference behind it may also apply to other similar objects and tasks, and stores it in a reusable form. Specifically, it uses an LLM to split the correction ζ into an object-agnostic *feedback template* f_ζ and an *anchor object* x_ζ, such that $f_\zeta(x_\zeta)$ is semantically equivalent to ζ. If ζ does not name an object, the LLM resolves x_ζ from the task context. To consolidate feedback, CO-OPT asks an LLM whether f_ζ is semantically identical to the template f_v of an existing entry $v \in V$ (Line 4); if so, ζ is merged into v, otherwise a new entry is created. Each entry v thus owns one unique template f_v and the representations of all anchor objects merged into it, referred as the *representation set* Φ_v.

Algorithm 1: Feedback Distillation

Input: Correction ζ, task T_i, database V
Output: Updated database V

1 $(x_\zeta, f_\zeta) \leftarrow$ ProcessFeedback(ζ, T_i)
2 $\{c_\zeta^1, \ldots, c_\zeta^M\} \leftarrow$ GenerateMCategories(x_ζ, f_ζ, T_i)
3 $\phi(x_\zeta, f_\zeta) \leftarrow \{\phi_{VL}(x_\zeta), \phi_{VL}(c_\zeta^1), \ldots, \phi_{VL}(c_\zeta^M)\}$
4 **if** $\exists v \in V : f_v = f_\zeta$ **then**
5 $\Phi_v \leftarrow \Phi_v \cup \{\phi(x_\zeta, f_\zeta)\}$
6 **else**
7 $v' \leftarrow$ NewEntry(f_ζ); $\Phi_{v'} \leftarrow \{\phi(x_\zeta, f_\zeta)\}$
8 $V \leftarrow V \cup \{v'\}$
9 **return** V

Representing Objects (Lines 2–3). Whether a stored template is retrieved for a new object depends on how its anchor object is represented. Following prior use of LLM-derived concepts to generalize feedback [3, 33, 34], CO-OPT prompts an LLM with x_ζ, f_ζ, and the task context for M object categories $c_\zeta^1, \ldots, c_\zeta^M$ capturing the intent behind the correction (e.g., *valuable items* and *electronics* for an cell phone). A vision-language encoder ϕ_{VL} then generates separate normalized embeddings for the object and each category and groups them to create $\phi(x_\zeta, f_\zeta)$ (Line 3).

3.2 Preference Retrieval (Alg. 2)

When the robot encounters an object x during a task, CO-OPT must estimate what the user would say about x, so that the planner can honor the preference before the user has to correct it. Retrieval produces this estimate from the stored templates by answering two questions: which of them apply to x, and which of them can be applied together. CO-OPT answers the first with a similarity measure over anchor representations and the second with a *conflict hypergraph*.

Calculating Similarity (Line 7). CO-OPT determines the relevance of a stored template to x based on x's similarity to the template's anchor object *and* the categories generated for it. The relevance of a stored representation $\phi(x_\zeta, f_\zeta)$ to x is thus the average cosine similarity between the normalized query embedding $\phi_{VL}(x)$ and the embeddings in the representation,

$$\mathrm{sim}\big(x, \phi(x_\zeta, f_\zeta)\big) = \tfrac{1}{M+1} \textstyle\sum_{a \in \phi(x_\zeta, f_\zeta)} \phi_{VL}(x) \cdot a, \tag{2}$$

following common practice in embodied knowledge retrieval [20]. An entry taught on several objects takes the relevance of its closest anchor: $\mathrm{sim}(x, v) = \max_{a \in \Phi_v} \mathrm{sim}(x, a)$.

Handling Conflicting Feedback (Lines 1–4). Standard vector retrieval returns the top-k entries most similar to x [19, 20]. For preferences, this fails in both directions: too large a k retrieves conflicting knowledge, and too small a k omits relevant preferences that do not conflict.

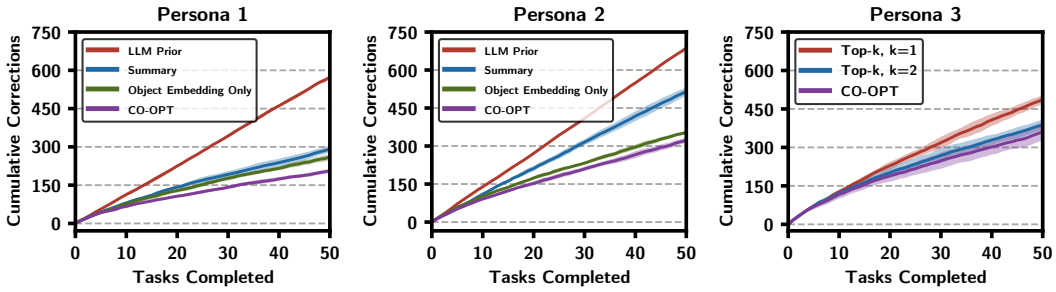


Figure 3: Cumulative correction counts over 50 tasks for three personas. The lines and shaded areas represent the mean and standard deviation respectively across 6 different randomly generated task sequences.

In LOOP, conflicting feedback applied to the same object may leave the planner no goal-reaching trajectory. Formally, a set of templates $e \subseteq V$ *over-constrains* π on task T_i if the preference estimate $p_e = \{f_v(x) : v \in e\}$ is infeasible (Sec. 2.1), i.e., the planner can no longer complete the task under it. CO-OPT performs this feasibility check (Line 3) by rolling out π under p_e and declaring a conflict when no rollout reaches the goal. CO-OPT tracks such sets as a *hypergraph* $H = (V, E)$ over V. Each entry is a node, and each hyperedge $e \in E$ is a set of templates that jointly over-constrain π on the same object. To keep conflict detection tractable, CO-OPT tests only subsets of size 2, reducing cost to $O(|V|^2)$. Additionally, we assume that conflicts *persist* across tasks, which allows us to lazily compute edges during retrieval.

Algorithm 2: Preference Retrieval

Input: Query object x, database V, conflict edges E, task T_i
Output: Object-oriented feedback p_x for x; updated E

1 **for** $e \subseteq V$ **do**
2 $p_e \leftarrow \{f_v(x) : v \in e\}$
3 **if** p_e *is infeasible for* T_i **then**
4 $E \leftarrow E \cup \{e\}$
5 **while** $|E| > 0$ **do**
6 $V' \leftarrow \{v \in V : \deg(v) > 0\}$
7 $v_{\min} \leftarrow \arg\min_{v \in V'} \mathrm{sim}(x, v)$
8 $V \leftarrow V \setminus \{v_{\min}\}$
9 $E \leftarrow \{e \in E : v_{\min} \notin e\}$
10 $p_x \leftarrow \{f_v(x) : v \in V\}$
11 **return** p_x

Retrieving Preferences (Lines 5–11). Given H and the query object x, CO-OPT iteratively removes the lowest-similarity node with degree greater than 0 along with its incident hyperedges until no hyperedge in the graph remains, guaranteeing that no conflicting feedback is retrieved. Finally, the templates of the remaining nodes are instantiated with x and returned as p_x. The total preference estimate p is then the union of p_x over the objects in the scene, $p = \psi(T_i) = \bigcup_{x \in O(s)} p_x$. p is then passed to the planner π to generate a preference-aligned plan. Although CO-OPT is not tied to a specific planner type, we use an LLM planner based on Code-as-Policies [35] in our work.

4 Experiments with Simulated Users

We evaluate CO-OPT's ability to generalize feedback over a task sequence longer than most prior evaluations [3, 9, 11, 36]. Since running such sequences with a physical robot is resource-intensive, we conduct these experiments in an abstracted simulated environment with simulated user personas.

4.1 Experimental Setup

Environment and Objects. Following the experimental approach of [3], we set up a text-world environment as an abstracted version of the domain in Sec. 2.3. A graph with the locations and receptacles as nodes serves as the navigation graph. To evaluate generalization in open-ended settings, we curate a set of 400 objects based on existing datasets [3, 37, 38]. These objects span 130 object categories split into 19 broader categories (e.g., dinnerware), 103 subcategories (e.g., utensils, plates), and 8 physical or semantic attributes spread across subcategories (e.g., fragile, valuable, sharp).

Simulated Users. To simulate users, we create three synthetic personas (see Fig. 2 right). **Persona 1** has high- and low-level preferences for the broadest object categories, while **Persona 2** has preferences over a mix of broad categories and more fine-grained subcategories. **Persona 3** has the same

broad-category preferences as Persona 1 but also includes additional non-conflicting high- and low-level preferences for each object. For each persona, we assign every object a set of non-conflicting preferences based on its categories. If the robot violates a preference for an object, the persona issues a correction, which the robot immediately distills. The robot then replans with a fresh set of retrieved feedback, and the stored correction remains available in subsequent tasks.

Protocol and Metrics. For each simulated run, we construct a sequence of 50 cleanup tasks, each containing 8 objects sampled without replacement from the dataset. Performance is determined by aggregating the number of corrections issued per task across the task sequence. We use Gemma 4 [39] as the LLM for both distillation and high-level planning, SigLIP 2 [40] as our embedding model ϕ_{VL}, and generate $M = 5$ object categories for our representation.

Baselines. We compare CO-OPT against four baselines. LLM-Prior, based on methods such as Code-as-Policies [35], does not store feedback and relies only on commonsense reasoning during LLM planning to infer preferences. Summary emulates LLM summarization-based methods such as TidyBot [3] by generating at retrieval time a two-paragraph summary of all the feedback received thus far and passing it to the LLM planner. ObjectEmbeddingOnly uses the same distillation and retrieval mechanisms as CO-OPT, but does not generate object categories. Top-k stores received feedback templates directly in its database without template consolidation and retrieves the top-k most similar templates using the same vector representation as CO-OPT, without considering feedback conflicts.

4.2 Experimental Results

Structured storage and retrieval of feedback reduces correction effort. Personas 1 and 2 test whether CO-OPT's structured storage and retrieval of feedback can reduce corrections (Fig. 3, left and middle). CO-OPT achieves the fewest cumulative corrections for both personas, reducing average total corrections by 64% and 53% (Personas 1 and 2) compared to LLM-Prior and by 29% and 37% compared to Summary. LLM-Prior never carries feedback forward, so it constantly receives corrections. Summary can handle preferences for broader categories but struggles to capture fine-grained preferences in its generated summary, listing specific objects rather than broad object categories. CO-OPT also improves over ObjectEmbeddingOnly; the larger gain on Persona 1 shows the benefit of generating object categories, especially for preferences over broader semantic groups.

Consolidating templates and tracking conflicts matters when preferences interact. Persona 3 isolates the utility of template consolidation and conflict tracking, since each object carries several non-conflicting preferences. We therefore compare CO-OPT against Top-k, which lacks both mechanisms (Fig. 3, right). Once again, CO-OPT achieves the lowest cumulative correction count, reducing average total corrections by 26% compared to Top-1 and by 7% compared to Top-2. Top-1 avoids conflicting feedback but leaves out the remaining non-conflicting preferences. Top-2 can in principle retrieve all non-conflicting feedback for an object. However, it can also retrieve two conflicting entries, in which case the planner may follow the wrong one, and without consolidation it can spend its budget on redundant entries and omit a critical one. In contrast, CO-OPT can flexibly handle any number of non-conflicting feedback instances for an object without needing to specify a specific k beforehand.

5 Conclusion and Future Work

In this paper, we formalized LOOP, the problem of learning object-oriented preferences over a lifelong sequence of tasks, and introduced CO-OPT, which learns such preferences from corrections while minimizing user effort. Our results indicate that *how* corrections are stored and retrieved, not only whether they are stored, is important to reducing user feedback effort. Since our framework was only tested in a simulated environment, future work will evaluate with human users in a real-world setting. By advancing lifelong preference learning, this work brings HRI systems closer to effective long-term deployment, in which an assistant that needs ever fewer corrections is most valuable.

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